ANGELHACK · BUILDING WITH IBM BOB · THEME 1 · STUDENTS / EARLY CAREER

Tomodoro Tune-Up

Taking an unfamiliar open-source app and, with IBM Bob Shell, learning how it works, fixing two real bugs and shipping the feature a student asked for.

David J · solo · student / early career

Bob Shell 2.0.6 on Linux, headless

7 Bob runs · ≈9 Bobcoins

0 → 29 passing tests

Live demo: **https://tomodoro-bob.sites.tab.bot/** · Repo: **https://gitlab.com/davidj-builds/tomodoro-tune-up** (public, MIT)

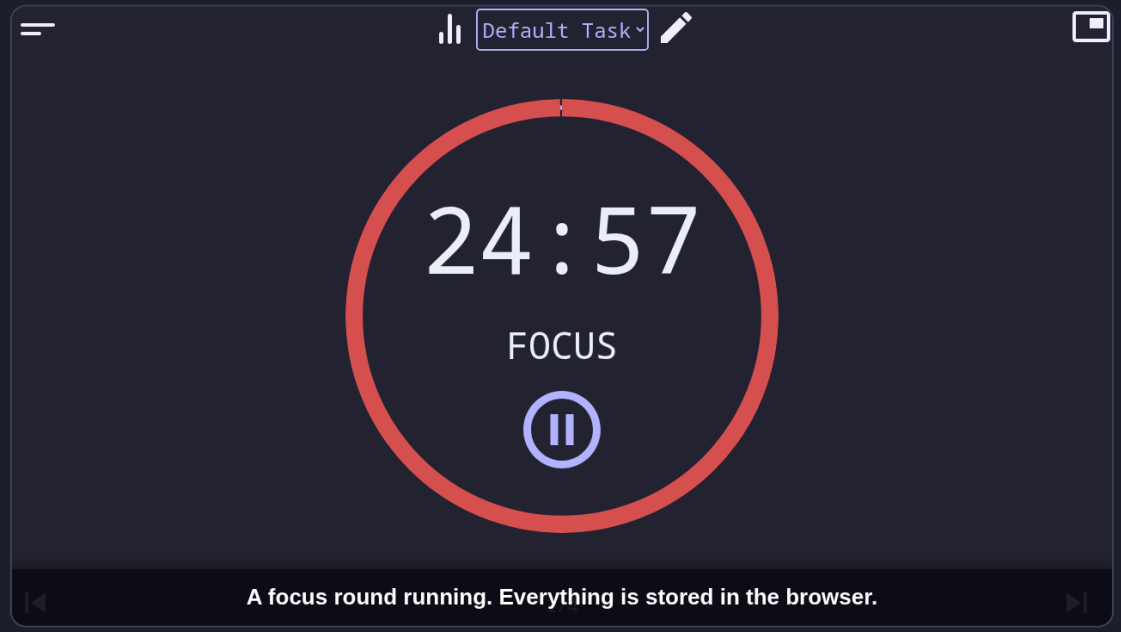
Disclosure: built with IBM Bob Shell, driven by David J with help from an AI assistant for prompting, checking Bob's claims and writing this deck. Every prompt and correction is in BOB-LOG.md.

THE STARTING POINT

An app I had never opened: lazy-guy/tomodoro

- Pomodoro web app: vanilla JS, ~3,700 lines, MIT, 300+ GitHub stars
- Tasks, statistics (pie + bars), picture-in-picture timer, white noise, offline via service worker + IndexedDB
- **No tests, no build step, one 1,800-line index.js**
- Real open issues from real users — #6 asks for per-subject colours in the pie chart: "I am a student… usually 7 subjects"

Baseline = upstream commit 17a5091. Everything after it was done during the hackathon (Oct 11, 2026).



One prompt → docs/ARCHITECTURE.md (368 lines, line-referenced)

```
$ bob run "I am new to this codebase... write docs/ARCHITECTURE.md explaining (1) what each file does;
  (2) the timer loop and autostart; (3) every localStorage/IndexedDB key; (4) how tasks, records and the
  statistics views are computed; (5) PiP and white noise; (6) places that look fragile or bug-prone,
  each with a file:line reference. Do not change any existing file."
→ 16 tool calls · 2m20s · 1.20 Bobcoins
```

Bob read every file and produced the map I worked from for the rest of the day, plus **nine** suspicious spots with line numbers.

I read each cited line before trusting it: 2 real bugs, 2 confident false alarms, 5 plausible UX notes.

```
§6.1 getRecords() resolves on last INDEX, not last completion  ✔ real
§6.7 restore: add() + no onerror → silent abort on duplicates  ✔ real
§6.2 nextRound() fires twice (worker sends maxDuration+1)      ✘ worker sends t-1
§6.5 maxH=1 so bucket 0 can never win the remark               ✘ loop uses v > 0
```

```
filteredArray.forEach((task, i) => {
  let tr
= ...index("task").getAll(IDBKeyRange.only(task));
  tr.onsuccess = () => {
    result.push(...tr.result);
    if (i === filteredArray.length - 1)
      resolve(result);   // last index ≠ last to
finish
  };
});
```

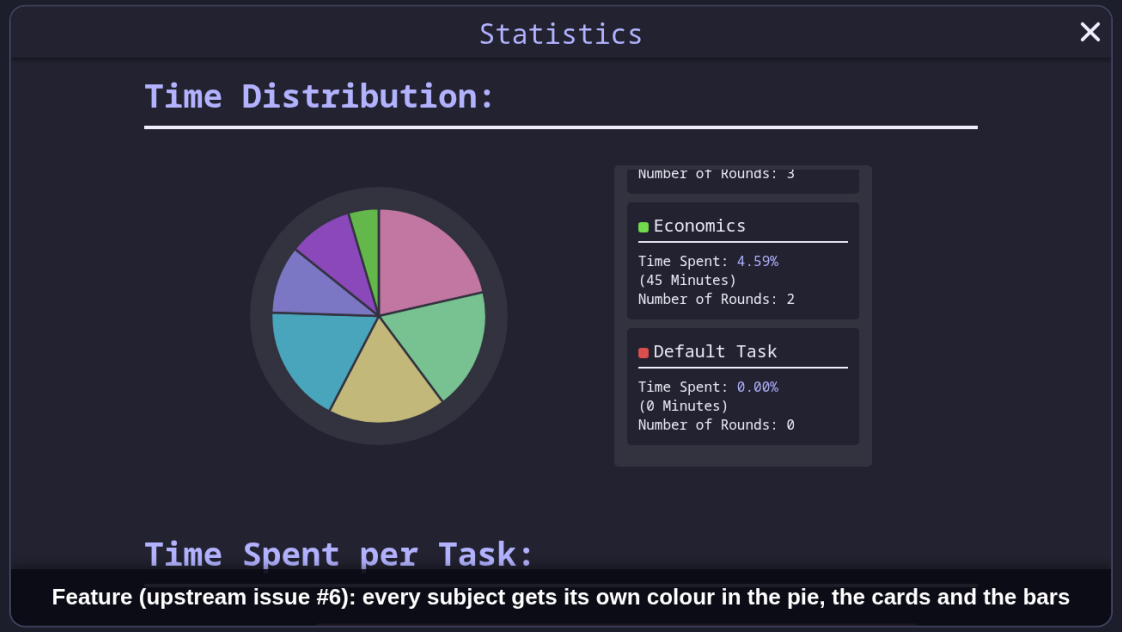
Filter statistics by two or more subjects: if the last request finishes first, earlier subjects' records are **silently dropped**.

Bob's fix (prompted test-first): `src/records.js` · `collectTaskRecords(store, tasks)` counts completions and resolves only when every request is in, in task order; rejects once on error.

Tests (node:test, a fake IndexedDB store with controlled completion order): one test re-implements the old algorithm and proves it drops records; six cover the fix, empty/null input and errors.

Second bug, same method: backup restore now validates entries (`normalizeBackup`, 15 tests), writes with idempotent `put()`, and tells the user what happened instead of failing silently.

Every subject gets its own colour: pie, cards and bars



Feature (upstream issue #6): every subject gets its own colour in the pie, the cards and the bars

- `src/colors.js · taskPalette(tasks)`: golden-angle hue spacing, alternating lightness, readable on dark and light themes, 12+ distinct colours (8 tests)
- Swatches added to the pie cards and the per-task bars so the legend matches the slice
- **Correction I made:** Bob keyed colours to the sorted slice order (as my prompt said), so a subject would change colour when rankings changed — one line changed to key by the user's stable task list
- Works offline: service-worker cache versioned (clock-v2) + activate cleanup so existing users get the new files

DEPTH OF ADOPTION — BOB AT EVERY STAGE, WITH RECEIPTS

Run	What Bob did	Kept / corrected	Cost
1 Explore	Architecture doc, 9 suspect areas	Kept; 2 findings disproved by reading the code	1.20
2 Fix	collectTaskRecords + fake-IDB tests, wiring, SW cache	Kept; test script fixed for Node 24	≈2*
3 Feature	colors.js, swatches, CSS, 8 tests	Kept; palette key changed to stable task order	2.70
4 Fix	normalizeBackup + 15 tests, idempotent restore, error UI	Kept	1.27
5 Review	Bob reviewed its own diff: 9 findings	3 applied, 1 disproved, 5 noted	1.17
6 Apply	SW versioning, settle-once reject, fix, 2 tests	Kept	1.03
7 Docs	README fork section	Kept; wrong test count fixed by hand	0.49

Every prompt, Bob's output, what was kept and what was corrected, dated, in **BOB-LOG.md**; raw transcripts in **docs/bob-runs/**. Workflow: explore → fix → build → fix → Bob reviews Bob → apply → docs. *Run 2's session ended before Bob printed its summary (see the log).

WHAT BOB GOT WRONG — AND WHAT THE ENVIRONMENT DID

- **"nextRound() can fire twice because the worker over-ticks"** — the worker posts t−1, so it fires once. Not touched.
- **"bucket 0 can never win the daily/hourly remark"** — the loop compares v > 0, so it can. Not touched.
- **Review: duplicate check breaks for d = "constructor"** — impossible, d must be a finite number first.
- **README claimed 17 backup tests** — there are 15. Fixed by hand.
- **Two Bob sessions were OOM-killed** on a 4 GB box when the pretty renderer met a git diff — worked around with JSON output and a saved diff file. Bob Shell also needs Node 24 (installed just for it) and its installer prompts on /dev/tty (has a --pm flag).

Habit that paid off every time: read the lines Bob cites before acting on them. Bob was right often enough to carry the project and wrong often enough to need a reader.

RESULTS & ROADMAP

Judging criteria, mapped

- **Tool usage & depth:** Bob in all 7 steps, logged, costed, corrected
- **Innovation & relevance:** a real app, a real user's request (#6), two real bugs — not a toy
- **Functionality:** live site, 29 tests green, offline cache migration handled, upstream-ready diff
- **Presentation:** 1:43 captioned demo, this deck, README, BOB-LOG

Next steps if continued

- Open the fixes as upstream pull requests (records race, restore, SW versioning)
- Finish issue #8 (notes on focus records) and #11 (spoken reminders) with the same test-first loop
- Pull the statistics maths out of index.js into tested modules (Bob's §6.5/6.9 notes)
- Use Bob's review mode as a pre-commit hook; keep BOB-LOG as the project's changelog

Live: https://tomodoro-bob.sites.tab.bot/ · Repo: https://gitlab.com/davidj-builds/tomodoro-tune-up · Upstream: github.com/lazy-guy/tomodoro (MIT, credited)